

Magnetic & Soft Medical Robotics

Shuwei Zhao

B.Eng. Candidate in Electronic Engineering
The Chinese University of Hong Kong

Design and experimental validation of miniaturized magnetic and compliant robotic systems for minimally invasive imaging and intervention.

- Miniaturized imaging catheters
- Magnetic actuation and navigation
- Soft robotic systems

SELECTED OUTPUTS

ICBIR 2026

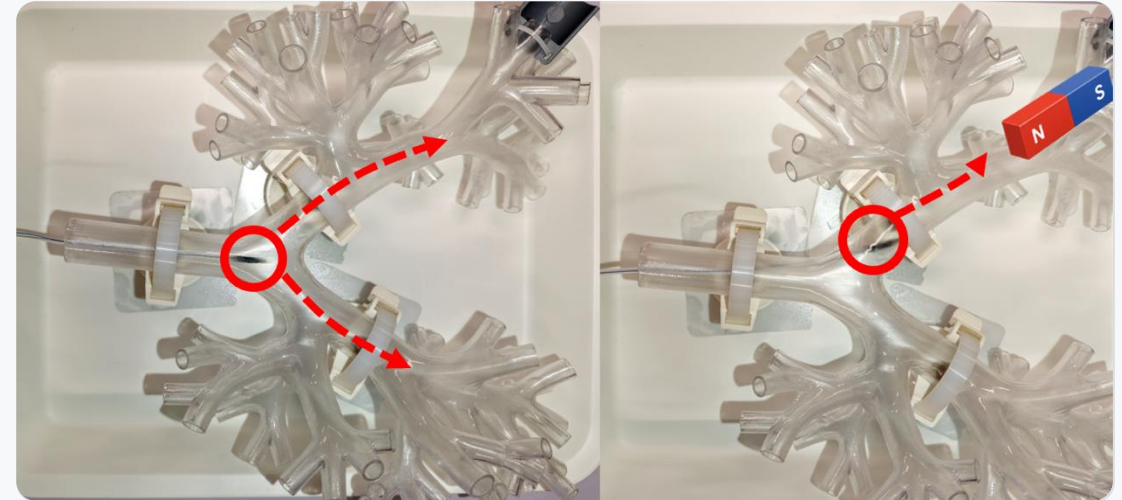
Co-first author

ICRA 2026

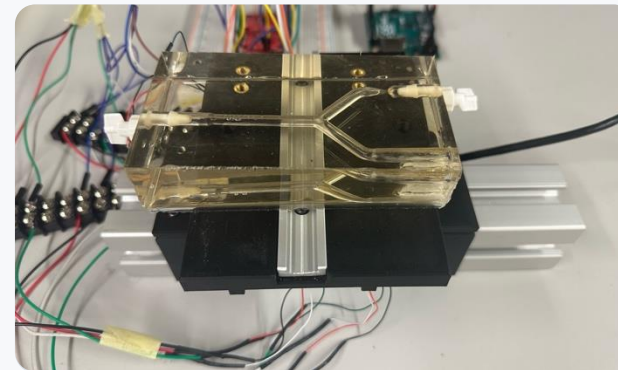
Accepted paper

ETH Zürich MSRL

Semester project · 5.5/6.0



Magnetic branch selection in a transparent bronchial-tree phantom



Quantitative characterization



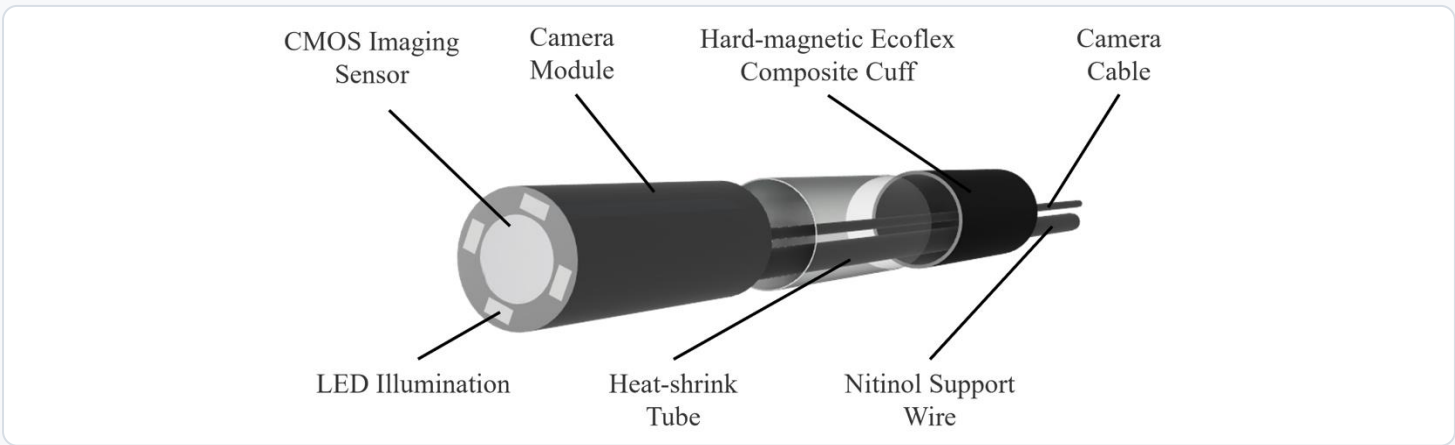
Control · integration · ex vivo validation

Magnetically Steerable Micro-Camera Catheter for Bronchial Navigation

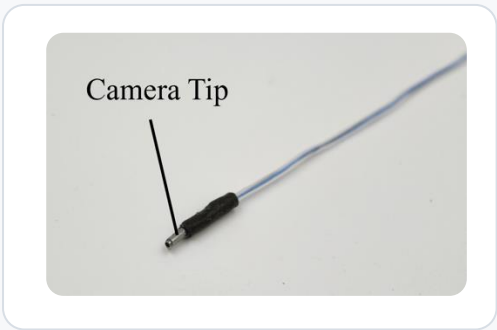
Co-first author · Accepted at ICBIR 2026 · CUHK · Advisor: Prof. Hongliang Ren

ENGINEERING PROBLEM Enable miniaturized forward-view imaging and controllable branch selection inside a bronchial-tree environment.

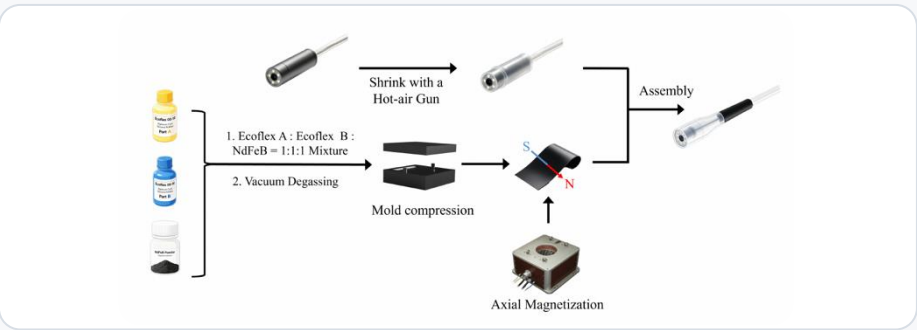
SYSTEM ARCHITECTURE



Camera + LED illumination · nitinol support · heat-shrink encapsulation · hard-magnetic composite cuff



Prototype



Composite preparation → magnetization → distal assembly

MY ROLE

01 DESIGN & FABRICATION

Designed and fabricated the millimeter-scale micro-camera catheter.

02 SYSTEM INTEGRATION

Integrated the camera, illumination, nitinol support, heat-shrink encapsulation, and hard-magnetic composite cuff.

03 EXPERIMENT DEVELOPMENT

Developed the insertion-force and steering-characterization setups.

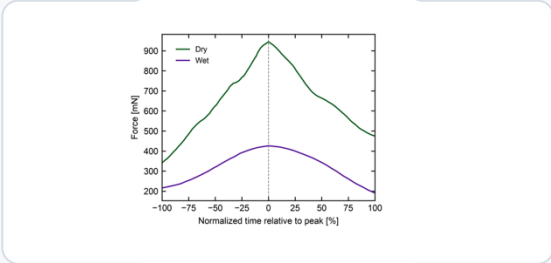
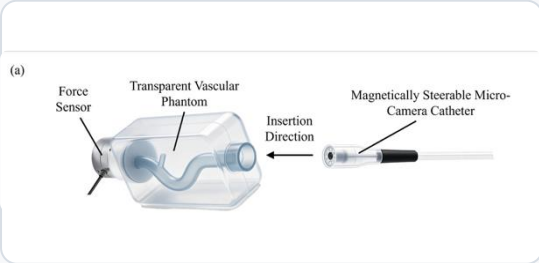
04 VALIDATION & DOCUMENTATION

Conducted bronchial-tree phantom navigation and contributed to manuscript and figure preparation.

Magnetic Steering Enabled Branch Selection While Preserving Forward Imaging

— QUANTITATIVE CHARACTERIZATION

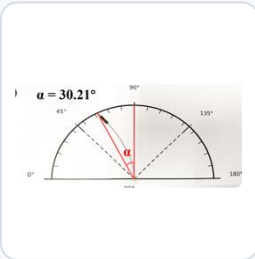
INSERTION FORCE



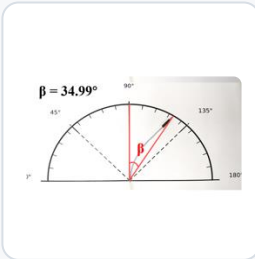
943.1 → 426.0 mN

Dry → saline-wetted · 10 repeated trials per condition

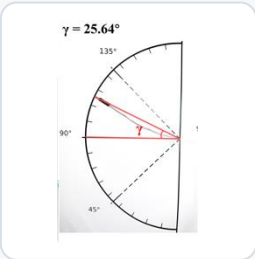
STEERING RANGE



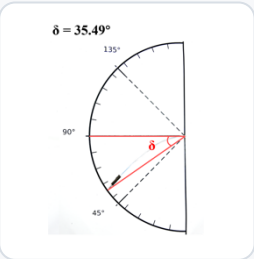
30.2°
left



35.0°
right



25.6°
up

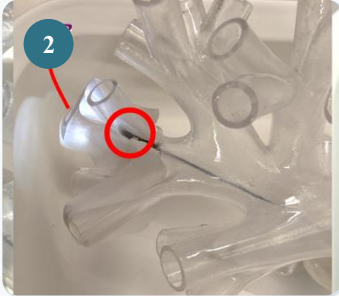
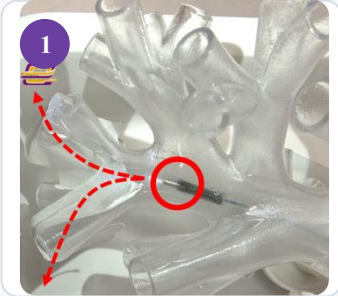


35.5°
down

Characterized insertion force and steering behavior.

— NAVIGATION AND IMAGING DEMONSTRATION

BRONCHIAL BIFURCATION SEQUENCE



FORWARD-LOOKING CAMERA VIEWS



Magnetic steering enabled branch-to-branch redirection in a bronchial-tree phantom.

Demonstrated redirection across bronchial-tree branches.

Maintained forward-looking imaging during benchtop navigation.

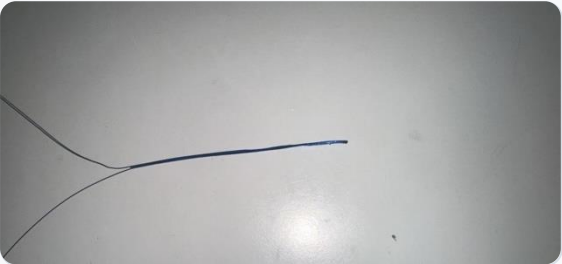
Soft Miniaturized Intraductal Imaging Catheter for Breast Duct Visualization

ETH Zürich · Multi-Scale Robotics Lab · Advisor: Prof. Bradley Nelson · Semester project grade: 5.5/6.0

DESIGN QUESTION

How can catheter diameter and insertion force be reduced while preserving sufficient pushability for intraductal imaging?

1 · DESIGN



0.5 mm and 0.7 mm nitinol supports

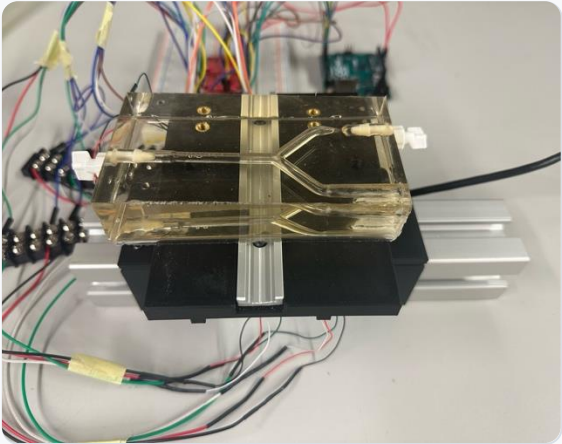


Soft phantom



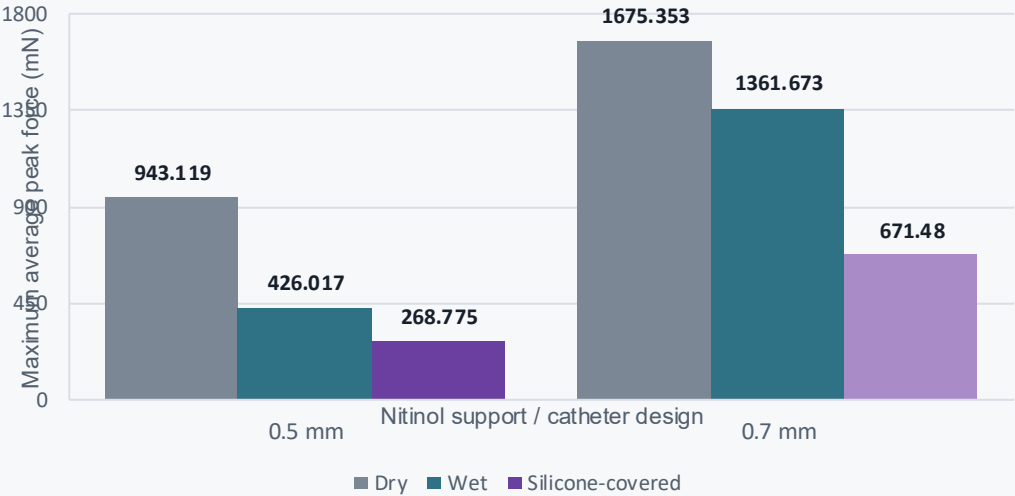
Camera view

2 · MEASUREMENT



Four load cells converted the catheter insertion event into a calibrated insertion-axis force measurement.

3 · RESULT



0.5 mm + silicone covering lowest measured force: 268.8 mN

10 selected trials analyzed per condition (20 manual trials recorded).

MY ROLE

Prototype design and fabrication

Force-platform development and calibration

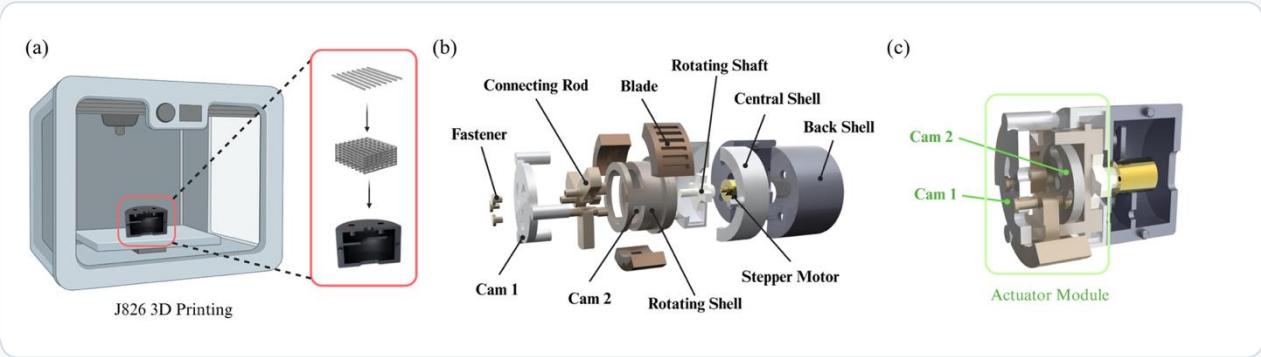
Experimental comparison

Phantom fabrication and imaging validation

Kirigami Capsule Robot for Minimally Invasive Gastrointestinal Biopsy

Contributing author · Accepted at ICRA 2026 · CUHK · Advisor: Prof. Hongliang Ren

— A · MECHANISM



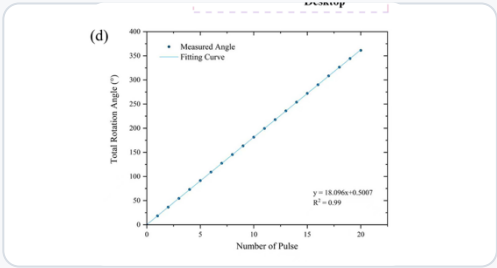
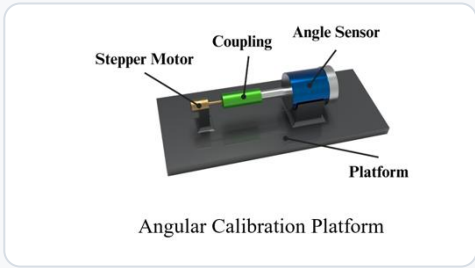
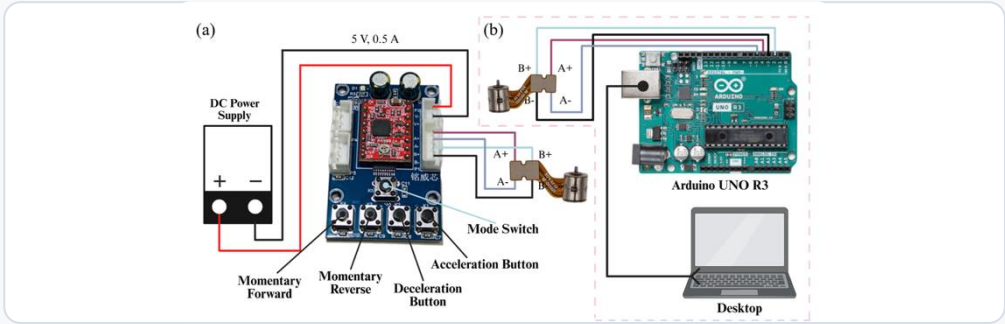
Exploded assembly clarifies the dual-cam actuator, rotating shaft, shell, and stepper-motor integration.

— C · ASSEMBLY AND VALIDATION



Ex vivo system validation

— B · CONTROL AND CALIBRATION



Contribution centered on control, calibration, integration, and validation.

$R^2 = 0.99$

MY CONTRIBUTION

- Developed motor-control electronics and Arduino pulse-control logic.
- Built and calibrated the angular-position measurement platform.
- Contributed to iterative mechanical assembly and system integration.
- Participated in ex vivo experiments and quantitative analysis.

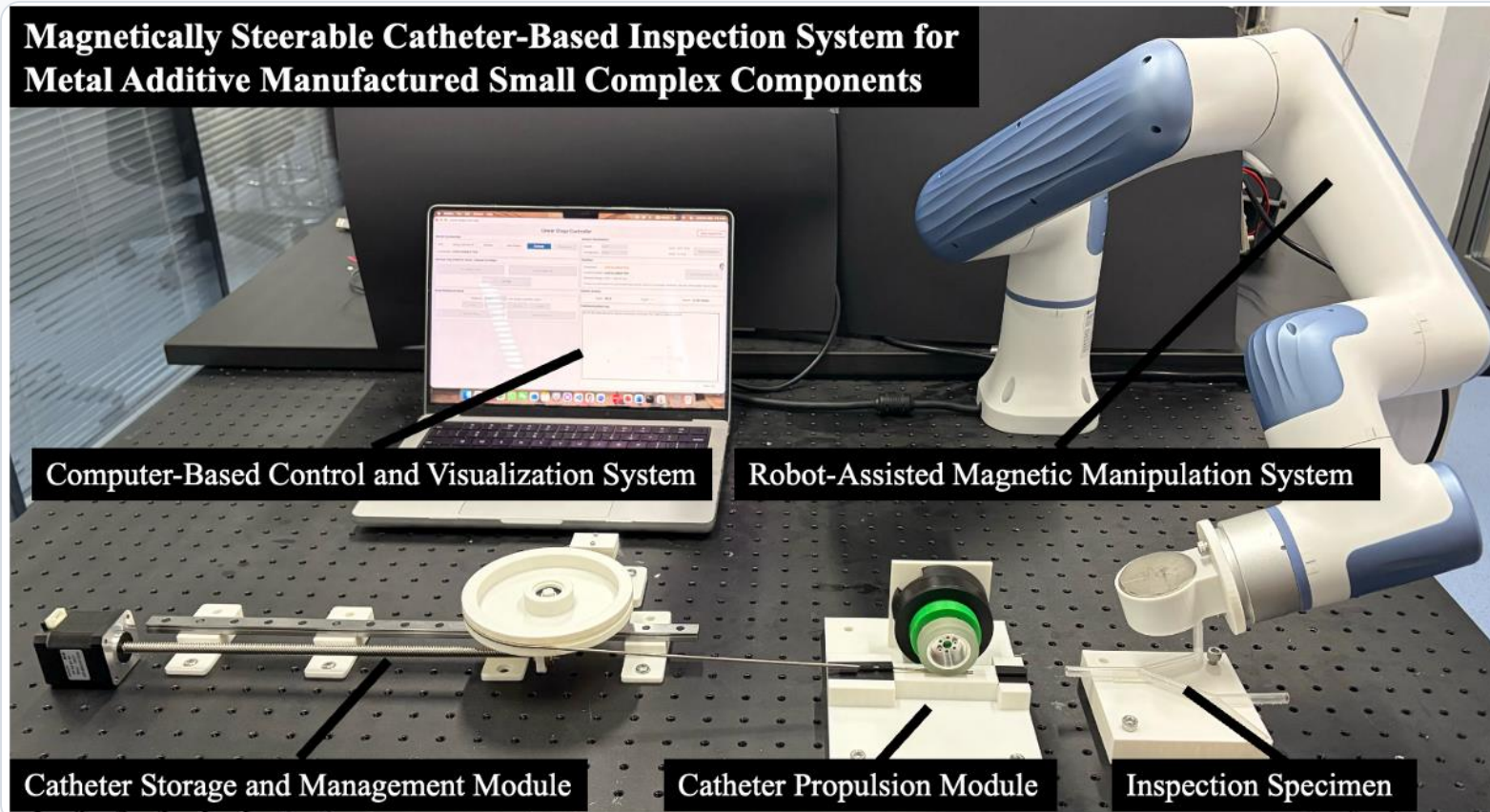
Motorized Catheter Storage and Management System

Magnetically steerable catheter system for industrial inspection · Xi'an Jiaotong University · Advisor: Prof. Qiji Ze

SYSTEM ROLE

Manage excess catheter length during insertion and retraction while preventing uncontrolled bending and entanglement.

Magnetically Steerable Catheter-Based Inspection System for Metal Additive Manufactured Small Complex Components



MY CONTRIBUTION

01 MECHANICAL SYSTEM

Designed and built the motorized translating-pulley catheter accumulator.

T8 lead-screw stage · MGN12H carriage
120 mm guide pulley · 3D-printed interfaces

02 PC CONTROL SOFTWARE

Developed the Python/Tkinter interface and the serial motion-control workflow.

Calibration · jogging · fixed-distance motion
Speed/acceleration settings · travel limits

$$\Delta L \approx 2\Delta x$$

2× length compensation